INCH-POUND

MIL-P-70507 (AR)
AMENDMENT 2
3 April 1989
SUPERSEDING
AMENDMENT 1
30 May 1986

MILITARY SPECIFICATION

PROJECTILE, 155MM, HE: XM864
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of Military Specification MIL-P-70507 (AR), dated 28 December 1984, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Add the designation "(M864)" in the title and change XM864 to "M864" wherever it appears in the specification.

PAGE 3

3.3: Delete in its entirety.

PAGE 5

4.3.1: Delete the following parts in their entirety:

"Expulsion charge assembly 9272019
Expulsion charge assembly 9362556".

Change the following drawing numbers:

"From 9362552 to 12630546
From 9362553 to 9391001
From 9362555 to 9391003".

Add additional first article items:

"Cup, Expulsion	9391057	25
Plate, Forward	9381111	10
Washer, Expulsion	12630555	25"

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PAGE 6

Table I.: Delete the following parts in their entirety:

"From 9362552 to 12630546
From 9362553 to 9391001
From 9362555 to 9391003".

PAGE 7

Table I.: Add the following new items:

"Washer, Expulsion (Dwg. 1263055)			
Examination for Defects	25	3.2	4.4.2.21
Cup, Expulsion (Dwg. 9391057)			
Examination for Defects	25	3.2	4.4.2.22".

PAGE 9

* 4.4.1: Delete the following restrictions:

g. "from one lot interfix number"

h. "from one lot interfix number"

Delete Item K and substitute the following:

"k. Expulsion charge container assemblies from one lot interfix number from one manufacturer."

Add the following item:

"o. Expulsion cups from one lot interfix number from one manufacturer."

PAGE 10

4.4.2.1: Delete in its entirety.

PAGE 11

4.4.2.2: Delete in its entirety.

PAGE 12

4.4.2.3: Delete in its entirety.

PAGE 13

4.4.2.4: Delete in its entirety.

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* 4.4.2.12: Delete Major Defect 101 and substitute the following:

"Major 101	Disassembly torque test of base burner assembly	100%	3.2	4.5.6"

Delete Major Defect 103 in its entirety.

PAGE 27

4.4.2.17: Change next drawing number from "9361552" to "12630546".
Change next higher assembly drawing number from "9362556" to "9391003".

PAGE 28

4.4.2.18: Change drawing numbers as follows:

From "9362553" to "9391001" and from "9362555" to "9391003"

Add major defects 110 and 111 as follows:

"110 Flatness of seal area	0.40%	3.2	Gage
111 Wall thickness	0.40%	3.2	Gage"

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4.4.2.19: Change drawing number "9362555" to "9391003".

PAGE 30

4.4.2.20: Change drawing numbers as follows:

"From 9362555 to 9391003"
"From 9362556 to 9381131"

Add the following major defects:

"105 Cup or lid damaged	0.40%	3.2	Visual
106 Gap between cup and lid 360°, max.	0.40%	3.2	Gage
107 Total length, max.	0.40%	3.2	Gage".

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4.4.2.21: Replace existing paragraph with new 4.4.2.21 as follows:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.21	Washer, Expulsion		1 OF 1	12630555
				NEXT HIGHER ASSEMBLY: 9391002

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	Angle of tape (two places 180° apart)		0.40%	3.2	Gage
Minor					
201	Poor workmanship		0.65%	3.2	Visual

NOTES:

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4.4.2.22: Replace existing paragraph with new 4.4.2.22 as follows:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.22	Cup, Expulsion		1 1		9391057
					NEXT HIGHER ASSEMBLY 9381131
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread		0.40%	3.2	Gage
102	Major diameter of thread		0.40%	3.2	Gage
103	True position of largest inside diameter with pitch diameter of threads		0.40%	3.2	Gage
104	True position of smallest inside diameter with pitch diameter of thread		0.40%	3.2	Gage
105	Minimum inside diameter		0.40%	3.2	Gage
106	True position of largest outside diameter with pitch diameter of thread		0.40%	3.2	Gage
107	Length of threaded area		0.40%	3.2	Gage
108	Hardness test (only applies for alternate material)	10	Note a	3.2	4.5.13
109	Thickness of bottom		0.40%	3.2	Gage
110	Thickness of sidewall		0.40%	3.2	Gage
111	Overall length		0.40%	3.2	Gage
112	Depth to start of angle		0.40%	3.2	Gage
113	Presence of two slots		0.40%	3.2	Visual
114	Largest outside diameter		0.40%	3.2	Gage
115	Perpendicularity of largest outside diameter		0.40%	3.2	Gage
116	Presence of angle between large and small inside diameters		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:
a. Each heat treat batch (see 6.7). If any cup fails to comply with the applicable requirement, the heat treat batch represented by the sample shall be rejected.

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4.4.2.24: Major 104:

Change "Concentricity" to "True Position".

PAGE 38

4.5.2: Delete paragraph 4.5.2 but retain 4.5.2.1.

PAGE 39

4.5.2.1: Change dwg. "936255" to "9391003".

PAGE 42

4.5.12: Delete the first part of the first sentence and substitute the following:

"After each sample of lid (dwg. 12630546) and empty cup (dwg. 9391001) have been assembled together for this inspection (dwg. 12630560), the assembly shall..."

Change drawing "(9362556)" to "(12630560)".

Add new paragraph 4.5.13 as follows:

"4.5.13 Hardness test of cup. The cups selected shall be subjected to the hardness test in accordance with the method and procedures of ASTM E 18-67. The load and penetrator shall be in accordance with the hardness scale specified on the applicable drawing. Three readings shall be taken on each cup on the outside surfaces of the closed and approximately 1/4" from the edge and approximately 120° apart. A suitable fixture shall be utilized to provide support under the surface being tested. Such fixtures shall be approved in accordance with 4.4.4. If the average value of the three readings for any cup is less than the value specified in the applicable drawing, that cup shall have failed the requirement. Equipment used for performing this test shall be approved in accordance with paragraph 4.4.4."

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Add new paragraph 6.7 as follows:

"6.7 Heat-treatment batch. A heat-treatment batch is herein defined as cups, of the same type, that have been produced from one lot of material, and heat treated at the same time in the same furnace, quench medium and aging process for each of all phases of the entire heat treatment process. When a continuous-type heat treatment process is used, the quantity of cups processed through the furnace for each four (4) hours production, or portion thereof, shall be considered as a heat treatment batch."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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